

From Agriculture to Services: Development without Industrialization

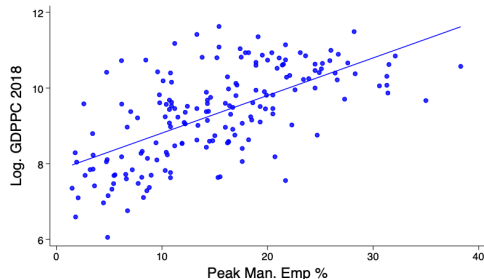
Tancredi Rapone

Bellaterra MacroClub, June 2026

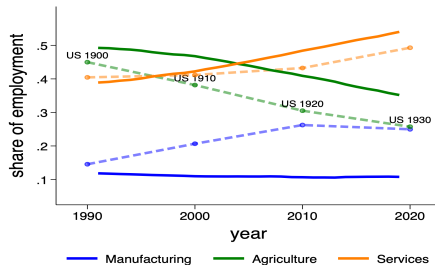
Barcelona School of Economics (BSE)

Universitat Autònoma de Barcelona (UAB)

Manufacturing Matters — But Many Countries Skip It



(a) Manufacturing peaks and income today

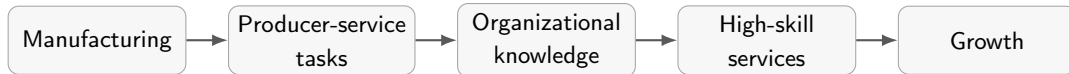


(b) Many developing countries skip manufacturing

Puzzle: Can countries close income gaps without industrializing?

Data: Maddison Project, ILOEST-Gapminder, GGDC.

This Paper: Manufacturing Shapes the Service Economy



Key mechanism

Manufacturing creates demand for high-skill producer services: cost control, quality control, logistics, inventories, management, and supplier organization.

Why it persists

Solving these problems creates portable routines and skills that later support high-skill producer services.

⇒ **Manufacturing leads to a high-skill service economy**

Historical Example: Managerial Accounting

**1914
DuPont**

manufacturing origin



ROI and financial control for complex industrial production.



**1922–24
McKinsey**

codification



Budgeting and managerial accounting become portable tools.



**1935
Marshall Field's**
McKinsey becomes CEO



Industrial control tools enter retail management.

Sources: DuPont financial control; McKinsey, *Budgetary Control* (1922), *Managerial Accounting* (1924); Marshall Field's, 1935.

► Across Countries Today

What I Do and What I Find

Document

Cross-country facts

Larger manufacturing peaks predict faster growth, more education, and lower fertility.

Identify

U.S. cross-county evidence

Manufacturing raises service skill intensity, school enrollment, and accelerates fertility decline.

Quantify

Unified Growth model

Manufacturing explains **60%** of GDPpc growth gaps, **53%** of education gaps, **39%** of population-growth gaps, and **31%** of income gaps.

Literature

- **Structural change and premature deindustrialization**

Kuznets (1966, 1973); Lewis (1954); Matsuyama (1992); Kongsamut et al. (2001); Ngai and Pissarides (2007); Herrendorf et al. (2014); Rodrik (2016); Comin et al. (2021); Sposi et al. (2021); Alfaro et al. (2023); Huneus and Rogerson (2024); Fujiwara and Matsuyama (2024).

⇒ I study the consequences of smaller manufacturing phases.

- **Service-led growth and production organization**

Buera and Kaboski (2012); Fan et al. (2023); Chen et al. (2023); Hsieh and Rossi-Hansberg (2023); Herrendorf et al. (2022); Duernecker and Herrendorf (2022); Rodrik and Sandhu (2025); Rodrik and Stiglitz (2025); Engbom et al. (2025); Peters et al. (2026).

⇒ I link manufacturing to the skill intensity of services.

- **Unified growth: fertility, education, and income**

Becker et al. (1990); Galor and Weil (2000); Hansen and Prescott (2002); Doepke (2004); Galor and Moav (2004); Galor (2005, 2011); Lagerlöf (2006).

⇒ I embed UGT in a model of structural change.

Roadmap

1. **Cross-country evidence**

Manufacturing peaks predict growth, education, service composition, and fertility.

2. **U.S. county evidence**

Historical manufacturing booms raise schooling, lower fertility, and shift services toward high-education occupations.

3. **Model**

Structural change with endogenous fertility, education, and heterogeneous producer services.

4. **Quantification**

Calibrate to the U.S. transition and vary the size of the manufacturing phase to account for cross-country patterns.

5. **Policy extension**

time permitting

Trade openness, producer-service subsidies, and manufacturing subsidies.

Cross-Country Evidence

Stylized Fact 1: Manufacturing GDP p.c. Growth Rates

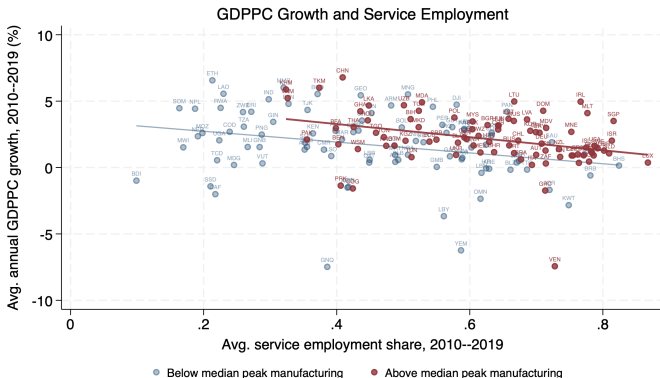


Figure 2: Baumol's cost disease. Countries with a higher peak of manufacturing employment continue to have higher growth rates. Data source: ILO and World Bank via Gapminder.

Stylized Fact 2: Manufacturing and Education

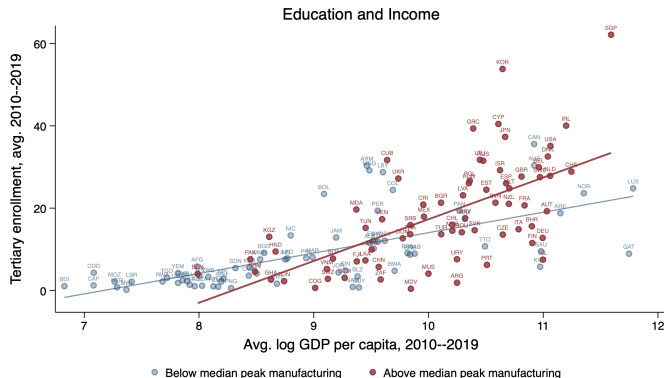


Figure 3: Tertiary enrollment rate and GDP per capita. Tertiary enrollments rise with development but more so if the economy goes through manufacturing. Data source: Barro and Lee (2015) and ILO via Gapminder.

Stylized Fact 3: Manufacturing and Fertility

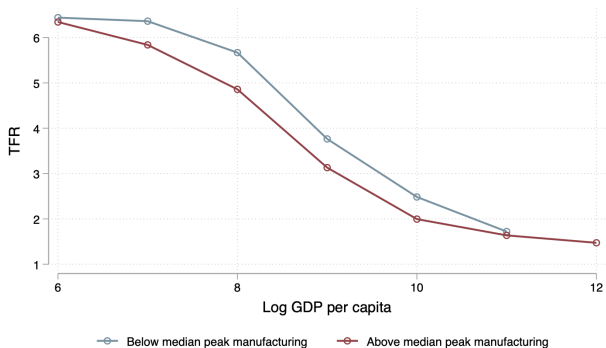


Figure 4: Scatterplot of the Total Fertility Rate (TFR) over log of GDP per capita across 176 countries observed from 1800 to 2020. Countries are grouped by being above and below median peak manufacturing employment share. Data Source: Gapminder.

Empirical Analysis

Difference-in-Difference:

$$EDSCOR50_{ct} = \delta_c + \gamma_t + \sum_{k \in \{-4, 6\}} \beta_k 1_{\% \text{Peak Man Emp.} \in \mathcal{P} \{t - T_{\text{peak}} = k\}} + \epsilon_{ct}$$

Long-Run Effect:

$$y_{c,s,2000} = \delta_s + \beta \text{Peak Man \% Emp}_{c,s,1850-1950} + \Gamma' X_{c,s,1850} + \epsilon_{c,s}$$

Note: $EDSCOR50_{ct}$ is education scores within services in county c and census decade t and 1 is a dummy variable equal to 1 if the county is k periods away from its peak, and its peak is in quantiles $\mathcal{P}$ of the cross county distribution.

DiD: Effect on Educational Scores Within Services

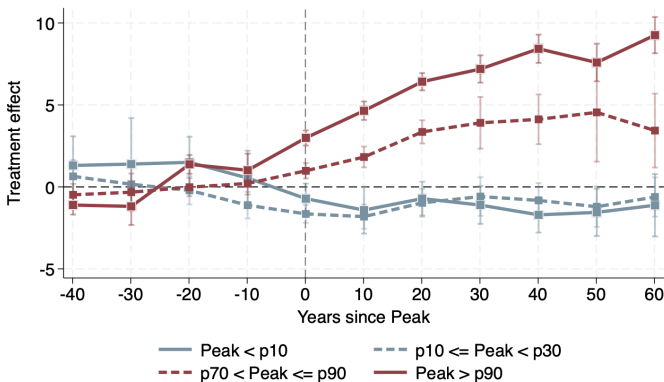


Figure 5: Specification: $Y_{ct} = \delta_c + \gamma_t + \sum_{k \neq -1} \beta_k 1\{\% \text{Peak Man. Emp.} \in \mathcal{P}, t - T_{\text{peak}} = k\} + \epsilon_{ct}$. Y_{ct} is the mean education score of service occupations in county c and decade t ; occupation scores equal the 1950 share with ≥ 1 year of college. Sources: U.S. Census (1850–1940); Haines (2010).

Decomposing the Effect by Service Occupation

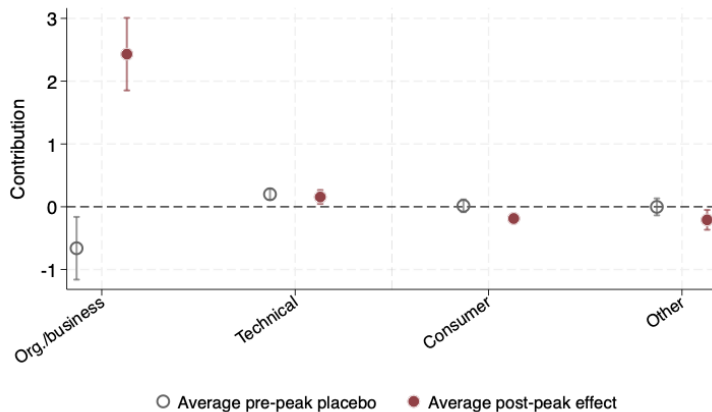


Figure 6: Occupational decomposition of the post-peak effect among service occupations excluding health, education, religion, and welfare. Data Sources: US Census of Population 1850-1940 and Manufactures 1880-1940.

Effect of Peak % Man. Emp. on Outcomes in 2000

Table 1: Effect of Peak % Manufacturing Employment on Outcomes in 2000

	Log patents pc	Frac. college+	Log income	Innovation / total emp.	HS services / total services
Panel A: OLS					
Peak Emp. % Man.	0.031*** (0.006)	0.127*** (0.036)	0.488*** (0.105)	0.055*** (0.021)	0.156*** (0.052)
State FE	Yes	Yes	Yes	Yes	Yes
Historical Controls	Yes	Yes	Yes	Yes	Yes
Observations	1600	1600	1600	997	997
R2	0.192	0.317	0.401	0.226	0.133
Panel B: IV					
Peak Emp. % Man.	0.042** (0.018)	0.421*** (0.156)	0.774 (0.471)	0.190* (0.099)	0.533* (0.283)
State FE	Yes	Yes	Yes	Yes	Yes
Historical Controls	Yes	Yes	Yes	Yes	Yes
Kleibergen-Paap	83.91	83.91	83.91	83.91	83.91
Observations	1600	1600	1600	997	997
R2	0.189	0.285	0.396	0.181	0.096

Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Model

Model Overview

Households

- Overlapping generations
- Adults work, consume, choose fertility
- Parents invest in child education
- Children become skilled or unskilled

Endogenous State variables

- Population: L_t
- Skilled share: s_t
- Sectoral productivity: $A_{j,t}$
- Organizational knowledge: H_t

Production

- Final sectors: $j \in \{a, m, c\}$
 - agriculture
 - manufacturing
 - consumer services
- Inputs:
 - value added
 - goods intermediates
 - producer services

Producer services

- Used by all final sectors
- Each task is low-skill or high-skill
- Tasks differ in codification hurdles

Household Problem

Parents in sector $i \in \{a, m, s\}$ choose consumption, fertility, and education:

$$\max_{\{c_{jt}\}, n_t, e_t} U_t = (1 - \gamma) \ln(C_t) + \gamma \ln(n_t h_t)$$

Consumption aggregator:

$$\sum_{j \in \{a, m, s\}} \omega_j^{\frac{1}{\sigma}} \left(\frac{C_{jt}}{C_t^{\varepsilon_j}} \right)^{\frac{\sigma-1}{\sigma}} = 1$$

Child human capital:

$$h_{it} = \frac{e_t + \tau_h}{e_t + \tau_h + g_t}$$

Budget constraint:

$$p_{at} C_{at} + C_{mt} + p_{st} C_{st} \leq (1 - n_t(\tau_i + \tau_e(S_t)e_t)) w_t(S_t)$$

► Skill choice

There are three final sectors: agriculture (a), manufacturing (m), and consumer services (c).

Value added in sector $j \in \{a, m, c\}$:

$$VA_{jt} = A_{jt} \left[\alpha_j (h_s L_{s,jt})^{\frac{\sigma_L - 1}{\sigma_L}} + (1 - \alpha_j) (h_u L_{u,jt})^{\frac{\sigma_L - 1}{\sigma_L}} \right]^{\frac{\sigma_L}{\sigma_L - 1}}$$

Gross output:

$$Y_{jt} = A_{go,jt} VA_{jt}^{\theta_{t,j}} M_{a,jt}^{\gamma_{a,t,j}} M_{m,jt}^{\gamma_{m,t,j}} PS_{jt}^{\gamma_{ps,t,j}}$$

$$\theta_{t,j} + \gamma_{a,t,j} + \gamma_{m,t,j} + \gamma_{ps,t,j} = 1$$

Producer-Service Composite

Producer services are used as an intermediate input in all final sectors.

Two producer-service modes:

$$HS_t = A_{hs,t} h_s L_{s,hs,t}, \quad LS_t = A_{ls,t} h_u L_{u,ls,t}$$

Sector-specific producer-service composite:

$$PS_{jt} = \left[x_{j,t} HS_t^{\frac{\sigma_{ps}-1}{\sigma_{ps}}} + (1 - x_{j,t}) LS_t^{\frac{\sigma_{ps}-1}{\sigma_{ps}}} \right]^{\frac{\sigma_{ps}}{\sigma_{ps}-1}}, \quad j \in \{a, m, c\}$$

Microfoundation of PS Inputs: Task Hurdles

In sector j , producer-service tasks differ in a codification hurdle $z \geq 0$.

Primitive cross-task distribution:

$$F_j(z) = 1 - e^{-z/\kappa_j}, \quad f_j(z) = \frac{1}{\kappa_j} e^{-z/\kappa_j}$$

Switching from LS to HS involved overcoming a hurdle with sector-specific distribution F_j

$$\rightarrow x_{j,t} = F_j(z_{j,t}^*)$$

Microfoundation of PS Inputs: Profit-Based Task Adoption

Producer-service tasks can be supplied in LS or HS mode.

LS mode unit cost:

$$c_t^{LS} = p_{ls,t}$$

HS mode unit cost for task z :

$$c_{j,t}^{HS}(z) = p_{hs,t} \exp\left(\frac{z}{\phi(H_t)} - 1\right)$$

A task is assigned to HS iff

$$c_{j,t}^{HS}(z) \leq c_t^{LS}$$

which implies the endogenous cutoff

$$z_{j,t}^* = \phi(H_t) \max\left\{1 + \log\left(\frac{p_{ls,t}}{p_{hs,t}}\right), 0\right\}$$

Result: Endogenous HS Task Shares

Given the sectoral hurdle distribution and the cutoff $z_{j,t}^*$,

$$x_{j,t} = \Pr(Z_j \leq z_{j,t}^*) = F_j(z_{j,t}^*)$$

So the HS producer-service task share in sector j is

$$x_{j,t} = 1 - \exp\left(-\frac{\phi(H_t) \max\left\{1 + \log\left(\frac{p_{ls,t}}{p_{hs,t}}\right), 0\right\}}{\kappa_j}\right)$$

$$p_{hs,t} = \frac{w_{s,t}}{A_{hs,t} h_s}, \quad p_{ls,t} = \frac{w_{u,t}}{A_{ls,t} h_u}$$

Given the distribution of producer codification hurdles, the corresponding hazard rate is constant:

$$h_j(z) = \frac{f_j(z)}{1 - F_j(z)} = \frac{1}{\kappa_j}$$

Hence one unit of frontier-oriented educated effort in sector j yields

$$q_{j,t} = \frac{1}{\kappa_j} s_{j,t} \bar{e}_t$$

Aggregating across sectors,

$$\tilde{H}_t = \bar{e}_t \sum_{j \in \{a, m, c\}} s_{j,t} \frac{1}{\kappa_j}$$

Population dynamics:

$$L_{t+1} = n_t L_t$$

Fertility by origin:

$$n_{i,t} = \frac{\gamma}{\tau_i + e_{i,t}}, \quad i \in \{a, m, s\}$$

Sectoral productivity growth:

$$\frac{A_{j,t+1}}{A_{j,t}} = 1 + H_t (s_t L_t)^{\phi_j}, \quad j \in \{a, m, s\}$$

Codification frontier:

$$H_{t+1} = H_t + \max\{\tilde{H}_t - H_t, 0\}$$

Proposition 1: Terminal service AGP. If final demand converges to services, the economy reaches a service-employment AGP.

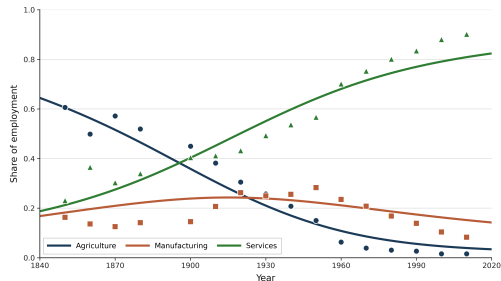
$$g^* = \frac{(\gamma - \tau_s + \tau_e(S^*)\tau_h)^2}{\tau_e(S^*)(\tau_s - \tau_e(S^*)\tau_h)} \quad (H^* \uparrow \implies S^* \uparrow \implies g^* \uparrow).$$

Proposition 2: Manufacturing vs. direct services. Branching experiment: 3 periods, same initial state; A goes through manufacturing, B goes directly to services.

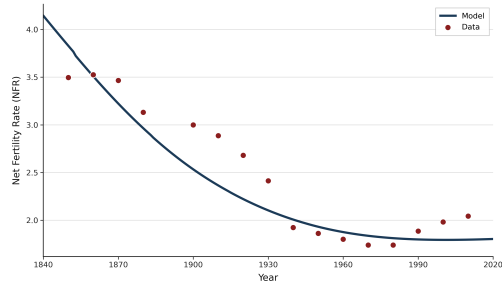
$$g_A^* > g_B^* \iff H_A^* > H_B^* \iff (s_1 L_1)^{\phi_m - \phi_s} \frac{\tau_m}{\tau_s} > \left(\frac{\kappa_m}{\kappa_s} \right)^2.$$

Calibration

Model Fit: SC and Fertility



(a) Structural change

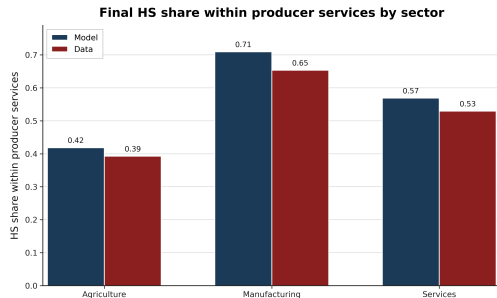


(b) Demographic transition

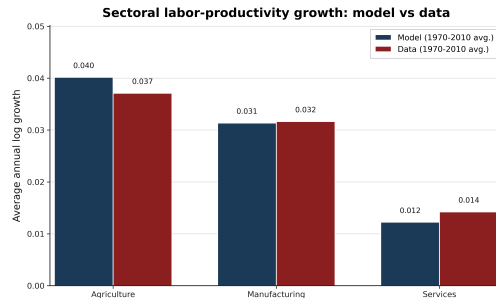
Figure 7: Model fit to targeted U.S. moments, 1850–2020. Dots represent data and lines show model simulations. The overall set of targeted moments includes: sector employment shares, sectoral productivity growth rates, sectoral and economy wide fertility rates, final high skill emp. share of producer services by sector, final skilled share and GDPPC levels. Data source: Gapminder.

► Full Table

Model Fit: Producer Services and VA Growth



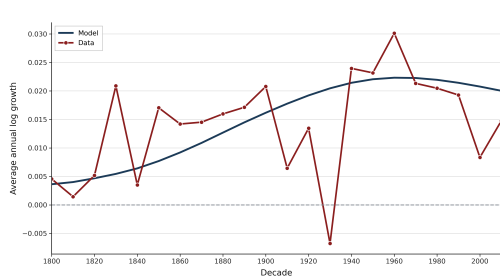
(a) Producer Services



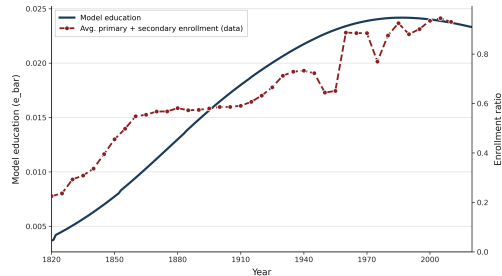
(b) Productivity

Figure 8: Model fit to targeted U.S. moments. The overall set of targeted moments includes: sector employment shares, sectoral productivity growth rates, sectoral and economy wide fertility rates, final high skill emp. share of producer services by sector, final skilled share and GDPPC levels. Data source: Gapminder.

Non Targeted Moments: GDPPC Growth and Schooling



(a) GDPPC growth



(b) Education vs. enrollment rates

Figure 9: Non-targeted validation moments: GDP per capita growth and schooling, 1800–2020. Schooling in the model is the average education choice across sectors and in the data is the average enrollment rate (primary + secondary)/2. Data source: Gapminder and US Censuses of Population.

Non Targeted Moments: The Effect of Peak Manufacturing Emp. Share

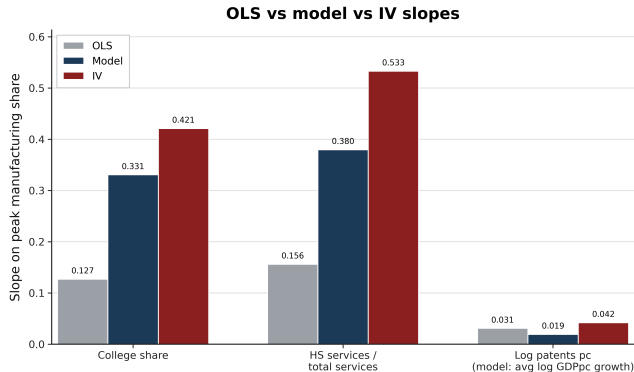


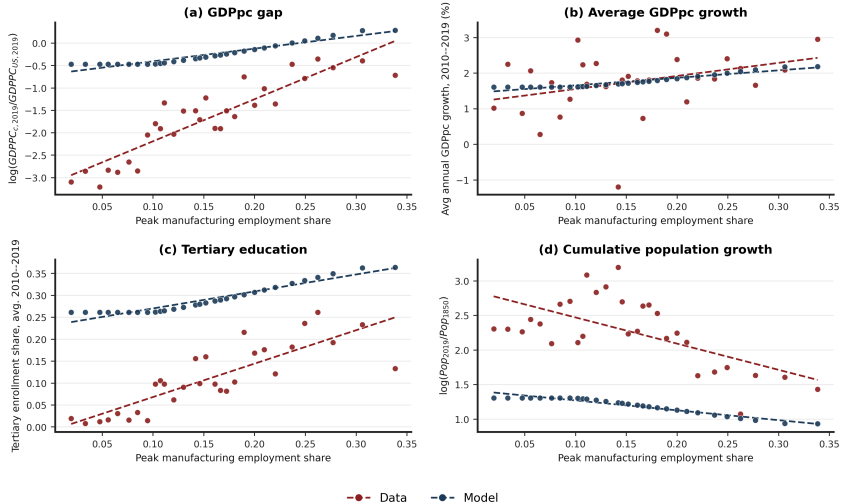
Figure 10: Comparison of the coefficients obtained from regressing key observable outcomes on the peak manufacturing employment share in the model and the data. Data slopes are consistent with Table 1.

Results

Questions we can ask the model:

- Can we reduce the US' peak manufacturing employment share? What happens to its growth rate, population and GDPPC today?
 - ⇒ Change ω_m , isomorphic to a final good tax on agriculture and services.
- How does the model account for the cross-country facts we observe in the data? These are
 - Positive relationship between a country's peak manufacturing employment share and its GDPPC growth rate.
 - Negative relationship between a country's peak manufacturing employment share and its population growth rate.
 - Positive relationship between a country's peak manufacturing employment share and its skilled share.

Accounting for Cross-Country Gradients



Taking Stock

Main idea: A structural-change model with endogenous fertility and education can account for key cross-country patterns.

⇒ Higher manufacturing peaks predict faster growth, more education, and faster demographic transitions.

Model exercise: Calibrate to the U.S. transition; vary manufacturing demand to generate economies with larger or smaller manufacturing phases.

Accounting results: The model explains:

- 29.7% of GDP per capita **level gaps**
- 58.0% of GDP per capita **growth gaps**
- 50.0% of tertiary education **level gaps**
- 39.1% of population **growth gaps**

Policy

Policy Exercise 1: Labor Subsidies

Question: Is industrial policy welfare improving? Which activities should be targeted?

Country economies

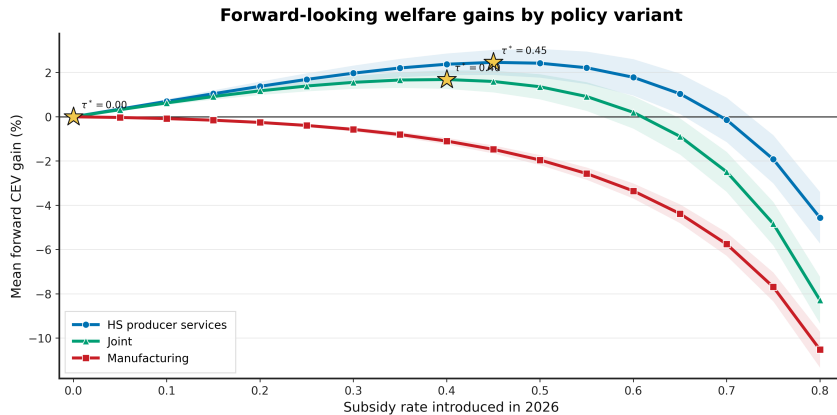
- Recalibrate ϕ_j , κ_j , and ω_m country-by-country
- Targets: GGDC-10s productivity growth, OECD-ICIO for HS shares, ILOEST manufacturing peaks (30 countries).

Policy experiment

- Permanent labor-cost subsidies from 2026 onward: $\tilde{w}_i^{firm,j} = (1 - \tau_j)w_i^{worker,j}$
- Financed with an income tax.

Evaluation: CEV through 2100

Optimal Subsidies: Manufacturing vs HS Producer-Services



Stars mark the mean-optimal subsidy rate for each policy variant.

What Creates Gains from IP? A Horse Race Approach

Exercise: Regress country-level CEV gains on all recalibrated country parameters.

$$CEV_c^P = \alpha^P + \beta_a^P \phi_{a,c} + \beta_m^P \phi_{m,c} + \beta_s^P \phi_{s,c} + \delta_a^P \kappa_{a,c} + \delta_m^P \kappa_{m,c} + \delta_c^P \kappa_{c,c} + \omega^P w_{m,c} + \varepsilon_c^P.$$

	Manufacturing	HS services	Joint
Agri. scale effect, ϕ_a	-0.005	-0.376**	-0.334**
Mfg. scale effect, ϕ_m	0.000	-0.163	-0.116
Serv. scale effect, ϕ_s	0.002	1.378***	0.948***
Agri. HS-task friction, κ_a	0.002	0.298	0.250*
Mfg. HS-task friction, κ_m	-0.004	0.548***	0.374**
Serv. HS-task friction, κ_c	-0.003	0.346	0.215
Mfg. demand shifter, w_m	0.001	0.147	-0.036
Observations	30	30	30
R^2	0.275	0.897	0.883

Standardized regressors; coefficients are percentage-point CEV gains from a one-s.d. increase.

Policy Exercise 2: Trade Openness

Environment

- Small open economy; Home trades agriculture and manufacturing.
- Consumer services and producer services are non-traded.
- Rest of world follows the same transition, shifted by a development gap Δ .

For traded goods $j \in \{a, m\}$, households consume an Armington composite:

$$p_{j,t}^A = \left[(1 - \lambda_j) p_{j,t}^{1-\theta} + \lambda_j (\tau p_{j,t}^*)^{1-\theta} \right]^{\frac{1}{1-\theta}}.$$

Counterfactuals

- **Open:** iceberg costs fall to 15% on opening date.
- **Protected:** same preferences, but prohibitive trade costs.
- Balanced trade each period:

$$\sum_{j \in \{a, m\}} M_{j,t} = \sum_{j \in \{a, m\}} X_{j,t}.$$

Trade Openness: Static Gains, Dynamic Costs

Trade gains by opening date and development gap

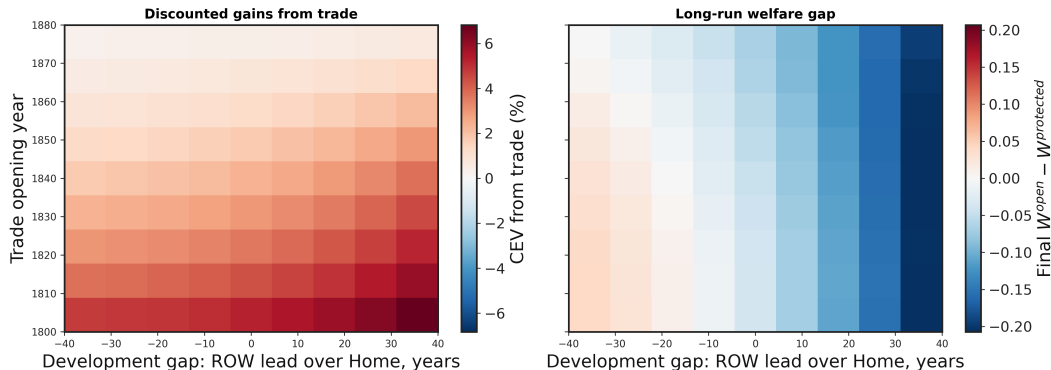


Figure 11: Discounted welfare gains from trade openness and long-run welfare gap.

Main finding: Manufacturing builds the capabilities behind high-skill services.

- Larger manufacturing phases predict faster growth, more education, lower fertility, and more high-skill services.
- The model explains 58% of GDPpc growth gaps, 50% of education gaps, and 39% of population-growth gaps.
- Manufacturing subsidies reduce welfare; high-skill producer-service subsidies raise welfare.
- Trade creates static gains, but can weaken long-run service-economy capability accumulation.

Appendix

Skill Formation

Skill choice probability:

$$p_{s,t} = \frac{1}{1 + \exp\left(-\frac{\beta_{\pi} \log \pi_t - c_0}{\sigma_{skill}}\right)}$$

where

$$\pi_t = \frac{w_{s,t}}{w_{u,t}}$$

Aggregate skill share dynamics:

$$s_{t+1} = \left(1 - \frac{1}{gen_t}\right) s_t + \frac{1}{gen_t} p_{s,t}$$

► Back

Motivation: Value Added

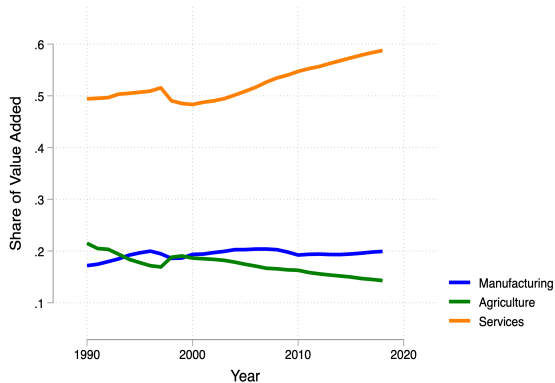


Figure 12: Value added shares over 1990-2018 in Low and Middle- income countries. Data source: ETD (2023).

DiD: Effect on Educational Scores Within Services (controls)

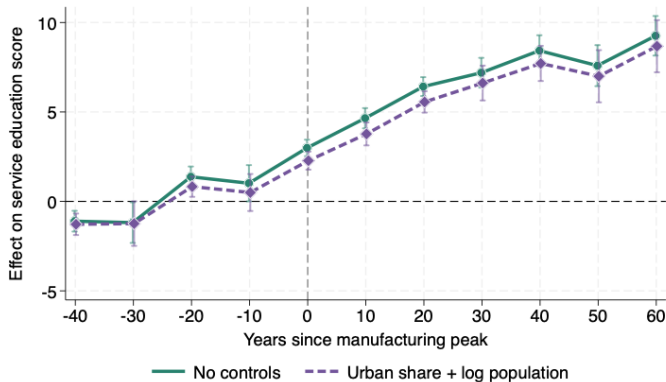


Figure 13: Specification: $Y_{ct} = \delta_c + \gamma_t + \sum_{k \neq -1} \beta_k 1\{\% \text{Peak Man. Emp.} \in \mathcal{P}, t - T_{\text{peak}} = k\} + \epsilon_{ct}$. Y_{ct} is the mean education score of service occupations in county c and decade t ; occupation scores equal the 1950 share with ≥ 1 year of college. Sources: U.S. Census (1850–1940); Haines (2010).

Accounting for Cross-Country Patterns

Table 2: Country-Level Model Fit

Outcome	N	Data	Model	Ratio	Corr.	R^2
GDP per capita	172	9.477	2.815	0.297	0.591	0.349
Tertiary enrollment	172	0.771	0.385	0.500	0.497	0.247
GDP per capita growth	172	3.603	2.091	0.580	0.122	0.015
Population growth	172	-3.633	-1.420	0.391	0.368	0.136
HS producer services	48	0.776	0.753	0.970	0.607	0.368

Notes: Data and model columns report slopes from separate regressions of each country-level moment on peak manufacturing employment. Ratio is the model slope divided by the data slope. Corr. and R^2 compare country-level model predictions with the data. All countries with non-missing data are included.

Calibration Fit

Calibrated parameter			Identifying moment		
Par.	Description	Value	Target	Data	Model
γ	Utility weight	0.043	Avg. fertility	2.474	2.351
τ_a	Fertility time cost, A	0.403	Fertility ratio A/S	2.000	2.031
ϕ_a	Scale effect TFP, A	0.985	TFP growth, A	0.037	0.040
ϕ_m	Scale effect TFP, M	0.884	TFP growth, M	0.032	0.031
ϕ_s	Scale effect TFP, S	0.547	TFP growth, S	0.014	0.012
g^{GO}	General TFP growth	0.003	Avg. GDPpc deviation	0.000	-0.126
κ_a	HS-task friction, A	1.136	HS in PS, A, 2019	0.393	0.416
κ_m	HS-task friction, M	0.490	HS in PS, M, 2019	0.653	0.706
κ_s	HS-task friction, S	0.726	HS in PS, S, 2019	0.529	0.566
η_H	HS frontier shape	4.729	Δ HS in PS, A	0.182	0.186
c_{col}	College cost	0.135	Skill share, 2019	0.345	0.325
α_a	Skill share VA, A	0.160	Within-VA skill, A	0.160	0.162
α_m	Skill share VA, M	0.225	Within-VA skill, M	0.220	0.228
α_s	Skill share VA, S	0.209	Within-VA skill, S	0.200	0.213
ϵ_a	Innovation efficiency, A	0.013	Avg. agri. share	0.259	0.249
ϵ_s	Innovation efficiency, S	1.082	Avg. serv. share	0.560	0.544
w_a	Utility weight, A	0.910	Agri. share, 1850	0.606	0.605
w_s	Utility weight, S	3.999	Serv. share, 1850	0.231	0.212

Fact 1: Growth Regression

Table 3: Dep. Var. is Avg GDPPC growth rate 2010-2019

	(1)	(2)	(3)	(4)
Avg. Serv. Emp. Share, 2010–2019	-2.742*** (-2.84)	-5.171*** (-4.49)	-7.566*** (-3.59)	-9.962*** (-4.17)
Peak Emp. % Man.		9.658*** (4.59)	5.455** (2.16)	5.511** (2.09)
N	174	174	174	174
R2	0.0469	0.130	0.215	0.284
Income Quintile FE	no	no	yes	yes
Region FE	no	no	yes	yes
Controls	no	no	no	yes

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Fact 1: Growth and Initial Income

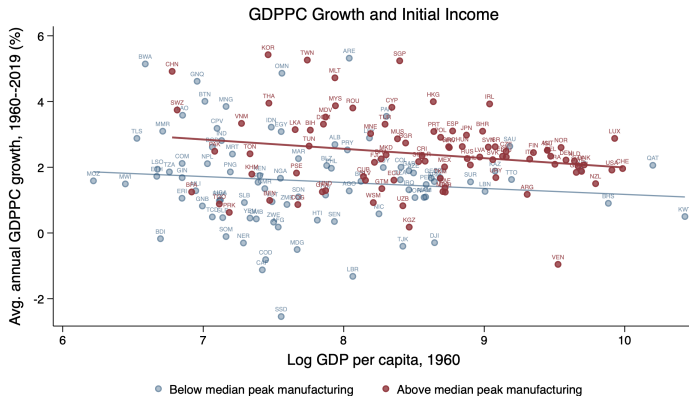


Figure 14: Average annual GDP per capita growth over 1960–2020 against initial GDP per capita in 1960. Countries are grouped by above- and below-median peak manufacturing employment. Data source: Gapminder.

Fact 2: Human Capital Quality

Table 4: Dep. Var. is Human Capital Quality, 2015

	(1)	(2)	(3)	(4)
Avg. log GDP per capita, 2010–2019	0.618*** (5.36)	0.523*** (2.74)	0.571* (1.82)	0.462 (1.42)
Peak Emp. % Man.	4.943*** (3.97)	4.866*** (4.05)	3.554*** (2.91)	3.237*** (2.66)
N	79	79	79	79
R2	0.552	0.559	0.697	0.702
Income Quintile FE	no	no	yes	yes
Region FE	no	no	yes	yes
Controls	no	no	no	yes

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Fact 2: Heterogeneity Within Services

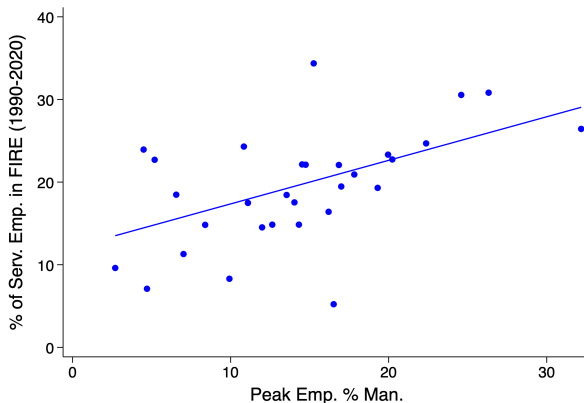
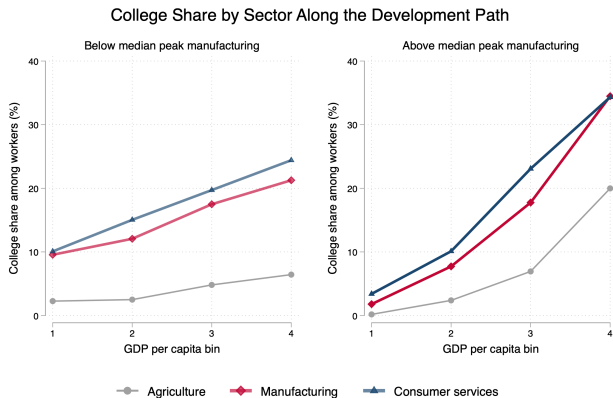


Figure 15: Binscatter of the fraction of service workers in high-productivity services against peak manufacturing employment. High-productivity services include FIRE, health care, and legal services. Data sources: IPUMS International and GGDC.

Fact 2: Sectoral College Shares over Development



GDPpc bins are common across country-years. Countries observed in only one bin are dropped. Countries weighted equally within sector-bin.

Figure 16: Average college employment shares by sector across GDP per capita bins, separately for countries below and above the median peak manufacturing employment share. Data source: IPUMS International.

Fact 2: Education Gains by Manufacturing Peak

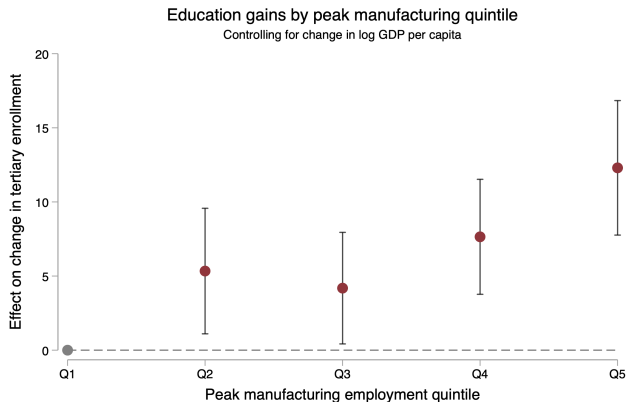


Figure 17: Coefficients from a country-level long-difference regression of the change in tertiary education on the change in log GDP per capita and peak-manufacturing quintile indicators. The omitted group is the lowest peak-manufacturing quintile. Data sources: Barro-Lee and Gapminder.

Fact 3: Fertility Regression

Table 5: OLS Regressions

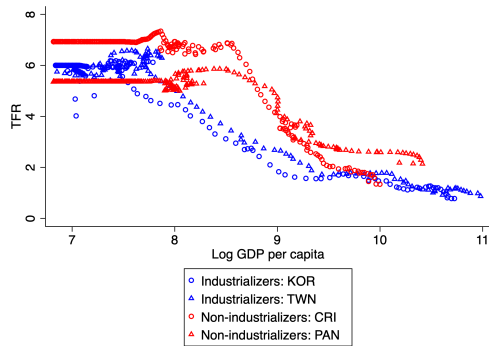
	(1)	(2)	(3)	(4)	(5)
% Man. Emp.	-13.43*** (-10.74)	-4.815*** (-4.85)	-4.815*** (-4.85)	-6.113*** (-5.40)	-6.826*** (-5.54)
Log GDP p.c.		-0.808*** (-5.58)	-0.808*** (-5.58)	-0.782*** (-5.33)	0.355** (3.29)
% Agr.		0.809 (1.07)	0.809 (1.07)	0.559 (0.72)	0.0945 (0.12)
N	5053	5053	5053	5053	5053
R2	0.326	0.655	0.655	0.676	0.966
Country FE	no	no	no	no	yes
Year FE	no	no	no	yes	yes

t statistics in parentheses

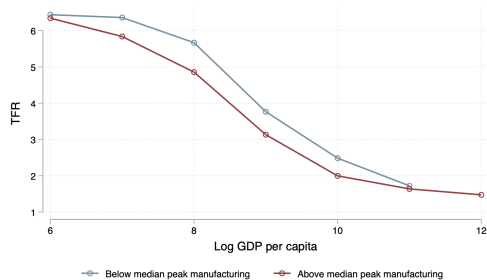
Dependent variable is total fertility rate. Standard errors are clustered at the country level.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Fact 3: Fertility and Development Paths



(a) Country comparison



(b) Above vs. below median peak

Figure 18: Total fertility rate over log GDP per capita. Data source: Gapminder.

Fact 3: Residual Fertility Model

To quantify fertility differences across sectors, I estimate:

$$nchild_{i,c,t} = \alpha_{a,c,t} + \delta_{e,c,t} + \gamma_{n,c,t} + \xi_{u,c,t} + \varepsilon_{i,c,t}$$

where $nchild_{i,c,t}$ is the number of children for woman i in country c and year t . The controls are country-year-age, country-year-education, country-year-non-working, and country-year-urban fixed effects.

The sample includes 46 million women aged 15–45 from 89 low- and middle-income countries.

Fact 3: Residual Fertility by Sector

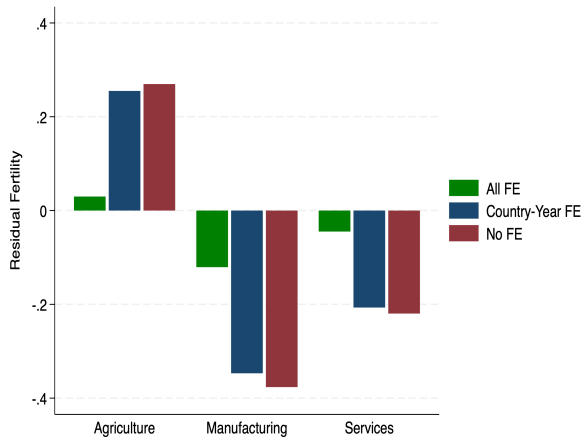


Figure 19: Average residual fertility by sector from the residual fertility model. Data source: IPUMS International.

Fact 3: Residual Fertility in the United States

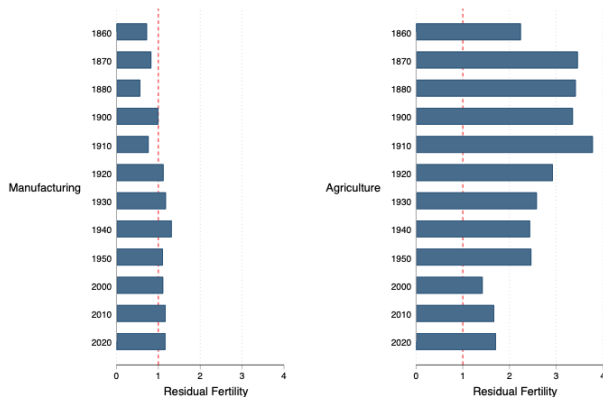


Figure 20: Sectoral fertility relative to services. Residuals are computed using samples of 25–35 year-old women from the U.S. Census of Population and normalized by service-sector fertility. Data source: IPUMS USA.

Fact 3: Self-Employment, Urbanization, and Fertility

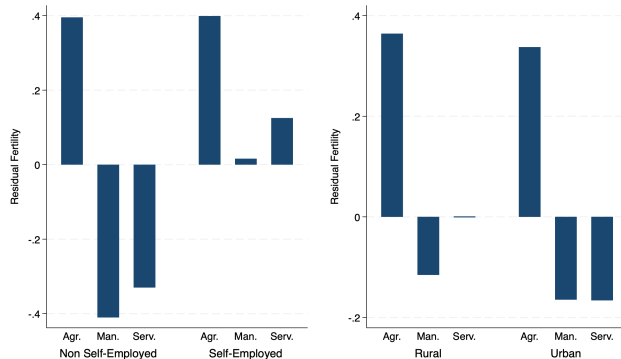


Figure 21: Residual fertility by sector, self-employment, and urban status. Residuals are from country-year fixed-effect regressions using women aged 25–45. Data source: IPUMS International.

Manufacturing Demand Pulls on High-Skill Producer Services

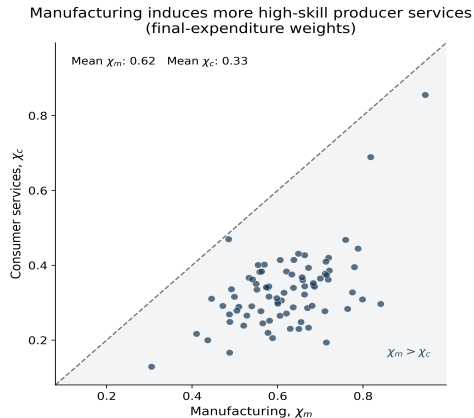


Figure 22: Each point is a country. χ_j is the high-skill share of producer-service output induced by a \$100 final-demand shock to sector j . The dashed line is the 45-degree line; points below it imply $\chi_m > \chi_c$. Consumer-service shocks are allocated using final-expenditure weights over market services. Data source: 2015 OECD ICIO.

Universities Across US Counties

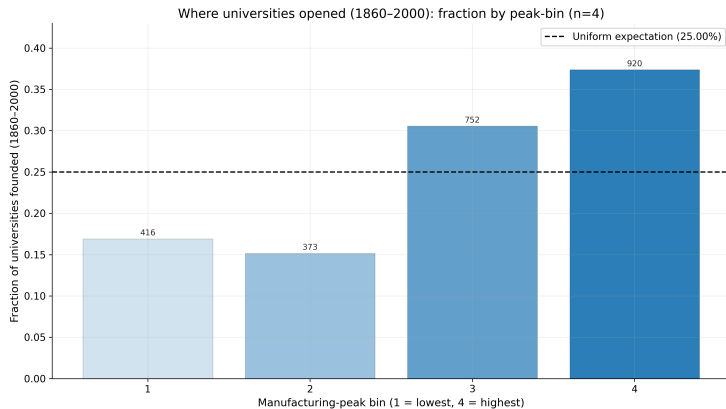


Figure 23: Fraction of universities founded between 1860 and 2000 across counties grouped into 4-quantiles of historical peak manufacturing intensity. Data on university openings is scraped from Wikidata.

DiD: Two Different Service Sectors (Industry Breakdown)

Bottom 10		Top 10	
Sub-industry	%	Sub-industry	%
Private households	3.5	Educational services	75.8
Shoe repair	4.3	Legal services	73.5
Trucking	5.3	Medical (excl. hospitals)	71.2
Taxicabs	5.7	Welfare & religious	59.1
Dressmaking	6.1	Engineering & architecture	58.4
Rail & bus transit	6.4	Misc. professional	53.9
Misc. repair	7.1	Drug stores	45.7
Sanitary services	7.2	Radio & TV	43.0
Auto repair	8.0	Accounting & auditing	42.7
Water transport	8.1	Hospitals	38.5

Note: Shares indicate the fraction of individuals in each sub-industry who completed one or more years of college, 1950 Census.

DiD: Two Different Service Sectors (Occupation Breakdown)

Bottom 10 (excl. manufacturing)		Top 10	
Occupation	%	Occupation	%
Teamsters	0.0	Mathematicians	100.0
Boatmen/canalmen/lock keepers	0.0	Biological scientists	96.0
Bootblacks	0.0	Chiropractors	93.9
Telegraph messengers	0.0	Professors/instructors (unspecified)	93.8
Midwives	0.0	Professors: Engineering	93.8
Hucksters & peddlers	1.7	Professors: Chemistry	93.8
Longshoremen & stevedores	2.1	Professors: Biological sciences	93.8
Laundresses (private household)	2.2	Professors: Mathematics	93.8
Charwomen & cleaners	2.7	Professors: Agricultural sciences	93.8
Private household workers (n.e.c.)	2.9	Professors: Physics	93.8

Note: Shares indicate the fraction of individuals in each occupation with ≥ 1 year of college (1950 Census). Manufacturing-heavy occupations (e.g., textile/metal/printing shop trades, mill/plant operatives, mining/industrial production jobs) are excluded from the bottom list.

DiD: Effect on Educational Scores (more groupings)

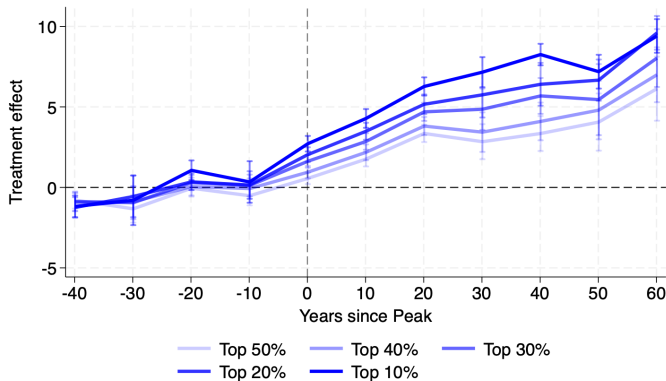


Figure 24: Specification: $Y_{ct} = \delta_c + \gamma_t + \sum_{k \neq -1} \beta_k 1\{\% \text{Peak Man. Emp.} \in \mathcal{P}, t - T_{\text{peak}} = k\} + \epsilon_{ct}$. Y_{ct} is the mean education score of service occupations in county c and decade t ; occupation scores equal the 1950 share with ≥ 1 year of college. Sources: U.S. Census (1850–1940); Haines (2010).

Map: Geography and the Peak

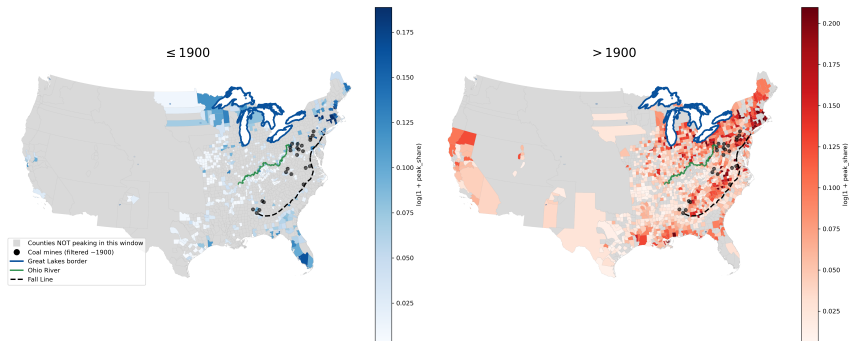


Figure 25: Map of peak manufacturing employment shares in the pre and post 1900 time periods. Data source: Author's calculations using Haines (2010), NHGIS and USGS.

IV: Instrumenting the Peak

The geographic features which are conducive to manufacturing change over time. For example, proximity to coal, rivers, or the Great Lakes may matter more in some decades than others.

I hence construct an instrument by taking the fitted values from the following zero stage regression:

$$\widehat{\text{Peak \% Man Emp.}}_i = \sum_d \left(\widehat{\beta}_d^C \ln(\text{coal})_i + \widehat{\beta}_d^F \ln(\text{fall})_i + \widehat{\beta}_d^O \ln(\text{ohio})_i + \widehat{\beta}_d^{GL} \text{GreatLakes}_i \right) \cdot 1\{\text{peak decade} = d\}$$

- Each geographic characteristic is allowed to have a different effect by peak decade.
- Identification comes from time-varying relevance of geographic advantages.

First Stage Results

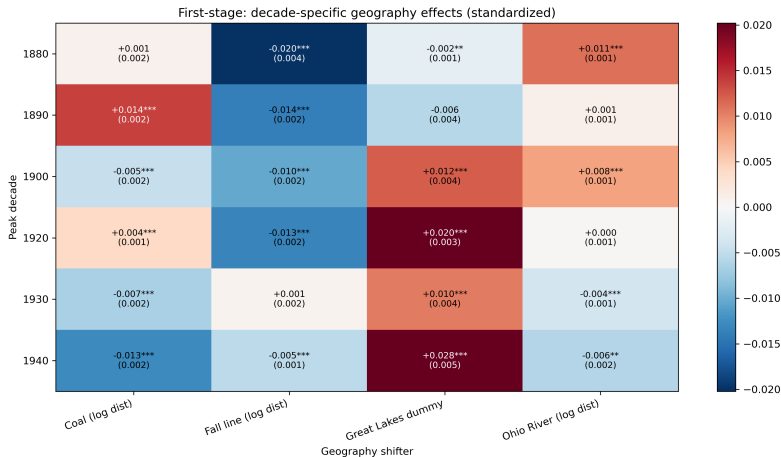


Figure 26: First stage (standardized) coefficients. Each geographic characteristic is interacted four decade peak dummy variables. Data source: US Census of population and Haines (2010).

DiD-IV: Effect on Educational Scores Within Services

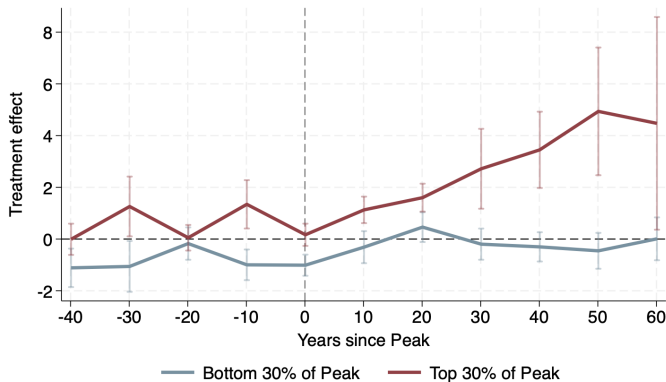


Figure 27: Specification: $Y_{ct} = \delta_c + \gamma_t + \sum_{k \neq -1} \beta_k 1\{\text{Peak \% Man. Emp.} \in \mathcal{P}, t - T_{\text{peak}} = k\} + \epsilon_{ct}$. Y_{ct} is the mean education score of service occupations in county c and decade t ; occupation scores equal the 1950 share with ≥ 1 year of college. Sources: U.S. Census (1850–1940); Haines (2010).

DiD: Constructing a Newspaper Labor-Demand Panel (1850–1940)

Step 1: Scrape universe of relevant pages (Chronicling America). Baseline data is from O'Rourke, Rapone and Santaaulalia-Llopis (2026), which uses a proximity search for “*help wanted*” (within 5 words) and downloads all $N \approx 550k$ pages from the LOC public repository.

Step 2: Build a stratified sample: I draw a *stratified random sample* at the newspaper level. $N \approx 110k$.

Step 3: Map newspapers to regions. I geo-locate each newspaper and cluster locations into $K = 200$ geographic regions using K-means on coordinates.

Step 4: Construct region-year outcomes from text counts. I count keyword hits for: (i) skilled manufacturing occupations (e.g., machinist, millwright, boilermaker, ...); (ii) skilled services/white-collar occupations (e.g., clerk, bookkeeper, stenographer, ...); (iii) a baseline basket: *cook + waiter + janitor + laborer*.

Step 5: Define treatment timing from “new/expanding factory” triggers. A region's treatment year is the first year with persistent mentions of new/expanding manufacturing establishments (threshold: ≥ 5 trigger pages in that region-year; restricted to 1880–1930).

Help Wanted Ads: Descriptive Results

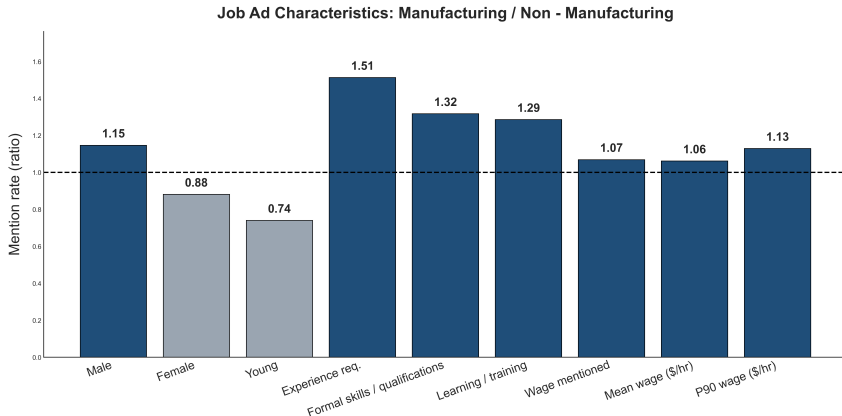


Figure 28: Job-ad characteristics in manufacturing vs. non-manufacturing contexts from historical U.S. newspapers (Chronicling America). An ad is classified as “manufacturing” if manufacturing context terms appear within ± 15 words; otherwise it is “general.” Each bar shows the relative mention rate of gender, age, experience/qualification and training related terms. When present, wages converted to hourly equivalents. Data source: Author’s calculations using LOC “Chronicling America”.

DiD: Event Study Effect of "New Factory" on Demand for Skilled Labor.

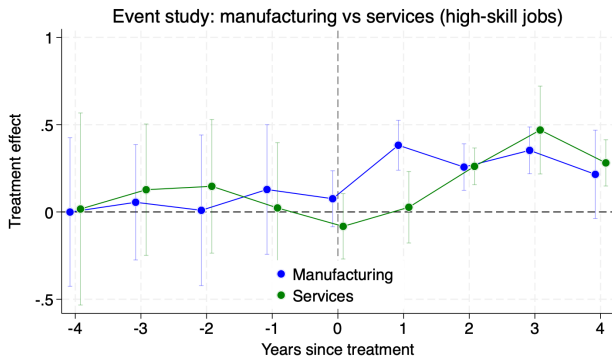


Figure 29: Event-study coefficients from Borusyak et al.'s imputation estimator using region-year newspaper data. The dependent variable is the log of occupation keyword hits normalized by a baseline basket of unskilled professions (cook, waiter, janitor, laborer). Vertical bars show 95% confidence intervals clustered by region. Data source: Author's calculations using LOC "Chronicling America".

DiD: Event Study Effect of "New Factory" on Demand for Skilled Labor Long Run

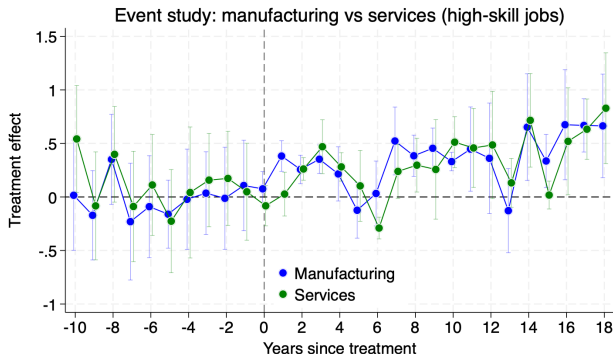


Figure 30: Event-study coefficients from Borusyak et al.'s imputation estimator using region-year newspaper data. The dependent variable is the log of occupation keyword hits normalized by a baseline basket of unskilled professions (cook, waiter, janitor, laborer). Vertical bars show 95% confidence intervals clustered by region. Data source: Author's calculations using LOC "Chronicling America".